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Roll No

AU-705 (GS)

B.E. VII Semester Examination, June 2020

Grading System (GS)

Computer Aided Engineering and FEM

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) How to model infinite degree of freedom into finite degree of freedom?
b) Explain the skeletal and continuum structures.
2. Explain the effect of node numbering on band width of banded matrix for
 - a) 1 D bar element
 - b) 2 D truss element
 - c) 3 D truss element
3. Write short notes on:
 - a) Effect of element aspect ratio on accuracy.
 - b) Numbering nodes for bandwidth minimization.
4. What does the term boundary condition mean and explain any one method for its application in structural engineering?
5. Derive the Solution of simultaneous equations using Numerical integrations.
OR
Describe the basic steps used in finite element problem formulations.
6. For a higher degree of freedom element explain the use of direction cosines [$1 = \cos\theta$ $m = \cos(90-\theta)$] in expressing the displacement vector in global coordinate system. Here θ is the inclination of the element from horizontal.
7. Define shape function for 2-D triangular elements.
8. Write a short notes on any two of the following:-
 - a) Hamilton's Principle
 - b) Use of natural co-ordinate system.
 - c) Banded and skyline assembly
